

GNN-Based BERT for Understanding Context from Music: A Hybrid Graph Neural Network and Language Model Approach

Shafin Ahmed

Dept. of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

shafin.ahmed6@g.bracu.ac.bd

Abstract—Music is a multi-layered signal with a “context” that encompasses genre, mood, harmony, timbre and listener-described semantics. Traditional sequence models based on spectrograms are able to identify local acoustic patterns, but fail to capture the relationship between the musical segments themselves across time. This paper proposes a hybrid model integrating Graph Neural Networks (GNN) with a pre-trained language model (DistilBERT) to capture the musical context on different levels of granularity. We present four increasingly complex tasks: (1) DistilBERT multi-label tag classifier on Google MusicCaps captions with Macro-F1 of 0.455 and AUC-PR of 0.616; (2) three-layer GraphSAGE encoder on segment-similarity graphs built from GTZAN audio compared with a 2D CNN mel-spectrogram baseline (55.0% vs. 63.0% accuracy); (3) cross-attention GNN-BERT fusion model for multimodal context prediction with AUC-PR of 0.337; and (4) InfoNCE contrastive dual-encoder for bidirectional text-to-audio retrieval with Recall@10 of 100%. We outline the entire pipeline, from preprocessing of audio data and construction of the graph, all the way to training and evaluation, and highlight the complementary capabilities of structural graph representations and contextual language embeddings for music understanding.

Index Terms—Graph Neural Networks, BERT, Music Information Retrieval, Multi-Label Classification, Cross-Modal Retrieval, Contrastive Learning, Audio Understanding.

I. INTRODUCTION

Music is more than just an audio wave. All these factors—genre and era, mood and emotion, harmonic progression, lyrical semantics, and timbral texture—are communicated in a single clip [1]. Music Information Retrieval (MIR) is still a major challenge in audio-only retrieval of all these contextual aspects. The classical methods have used Convolutional Neural Networks (CNNs) to mel-spectrogram representations [2], [3] or Recurrent Neural Networks (RNNs) to frame features in a sequence [4]. These methods work well at capturing the acoustic patterns of the local music, but do not explicitly model the relational structure of the music itself: how temporally distant segments share harmonic content, how chord progressions form cyclic patterns, or how repeated motifs create a structural skeleton for a composition.

Graphs provide a natural representation of relational data, which is well captured by Graph Neural Networks (GNNs) [5], [6]. A GNN can be used to propagate information along

musically meaningful pathways by constructing a graph with nodes representing audio segments and edges representing the adjacency in time or similarity in timbre. Meanwhile, large pre-trained language models like BERT [7] and DistilBERT [8] have shown amazing performance in capturing the semantic meaning of the text in the context, which could prove useful in processing text about music, like captions, tags, and listener descriptions.

The aim of this project is to create a hybrid system based on a GNN-BERT architecture to combine structural audio representations with language contextual embeddings. In line with the course project specification, we will employ four tasks of increasing difficulty:

- **Task 1 (Easy):** Multi-label tag classifier on natural-language captions of MusicCaps [9] using a DistilBERT model.
- **Task 2 (Medium):** An encoder based on GraphSAGE [6] over segment graphs built from GTZAN [10] audio compared to a CNN baseline on mel-spectrograms.
- **Task 3 (Hard):** Learning multimodal tags using a cross-attention fusion model between graph embeddings of GNNs and BERT hidden states.
- **Task 4 (Advanced):** An InfoNCE [11] contrastive dual-encoder, aligning representations of audio graph and text caption in a shared embedding space for bidirectional retrieval.

The rest of this paper is structured as follows. We discuss related work in Section II. The problem is formalized and the datasets described in Section III. The methodology is described in Section IV, which covers graph construction, model architectures and training procedures. Experimental results are given in Section V. Analysis and discussion are included in Section VI. This paper ends in Section VII.

II. RELATED WORK

A. Audio-Based Music Classification

Music classification is a well-studied subject in deep learning. Choi et al. [2] showed that CNNs trained on mel-spectrograms can yield good results for music tagging. Pons

et al. [12] investigated musically motivated CNN filter shapes for the aim of capturing timbral and temporal patterns. On the MagnaTagATune dataset, Won et al. [3] compared a variety of CNN and sample-level architectures for music tagging. Recently, attention-based models and transformers have been used for audio spectrogram classification [13]. But these methods are based on a grid or sequence of audio signals that do not explicitly indicate the structural relationships between sections of a musical piece.

B. Graph Neural Networks for Audio

There has been a number of applications of GNNs in audio tasks. For structured temporal data, Yan et al. [14] leveraged GNNs to tackle the action recognition task, showing the effectiveness of graph-based representation for such data. In the music field, graph representations have been investigated from the perspective of chord recognition and harmonic analysis. To learn node embeddings by sampling and aggregating features from local neighborhoods, Hamilton et al. [6] proposed GraphSAGE. Velickovic et al. [15] proposed Graph Attention Networks (GATs), which use attention mechanism for neighbor aggregation. We choose GraphSAGE for its scalability and the ability to learn from the inductive learning capability when used with music segment graphs.

C. Pre-trained Language Models for Music

BERT and its distilled version DistilBERT are now commonly used for text understanding. Language models have been used in music for applications such as music captioning, music tag prediction based on textual descriptions and lyrics understanding. The MusicCaps dataset [9] is a collection of 5,521 music clips with expert-written natural-language captions for music understanding research with text. These captions are encoded using DistilBERT for multi-label classification task and used as one arm of a cross-modal retrieval system.

D. Multimodal Fusion and Contrastive Learning

Contrastive objectives such as CLIP have been investigated for cross-modal learning between audio and text [16]. CLAP (Contrastive Language-Audio Pretraining) [17] trains a shared embedding space between the audio and the text, using the contrastive loss. Various multimodal fusion approaches can be used, such as simple concatenation or attention-based approaches [18]. We propose our cross-attention fusion (Task 3) using a formulation similar to cross-attention mechanisms in vision-language models [19] by using graph embeddings to query text hidden states. In Task 4, we use the InfoNCE objective [11] using in-batch negatives, which has been shown to be successful for learning aligned multimodal representations.

III. PROBLEM FORMULATION AND DATASETS

A. Problem Formulation

Let a music track be represented as a tuple $T = (X_{\text{audio}}, X_{\text{text}}, G, y)$ where X_{audio} denotes the raw audio signal (resampled to 22,050 Hz), X_{text} is the tokenized textual

description, $G = (V, E)$ is a segment graph with nodes V representing audio segments and edges E encoding temporal and similarity relationships, and y is the set of context labels (genre, tags, or aspect annotations).

A **BERT text encoder** maps text tokens to contextual embeddings:

$$H_{\text{text}} = \text{BERT}(X_{\text{text}}) \in \mathbb{R}^{L \times d} \quad (1)$$

where L is the sequence length and $d = 768$ is the hidden dimension.

A **GNN encoder** with K layers updates node features via message passing:

$$h_i^{(k+1)} = \sigma \left(W^{(k)} \cdot \text{AGG} \left(h_i^{(k)}, \{h_j^{(k)} : j \in \mathcal{N}(i)\} \right) \right) \quad (2)$$

where $h_i^{(0)}$ is initialized from the 160-dimensional audio segment feature vector.

B. Datasets

1) *GTZAN Genre Collection*: GTZAN has 1,000 audio tracks (30 seconds each) equally distributed over 10 genres: blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, and rock [10]. The number of tracks used are 999 (excluding one corrupted file) and the ratio used is 80/10/10, resulting in 799 tracks for training, 100 tracks for validation, and 100 tracks for test.

2) *Google MusicCaps*: MusicCaps [9] offers 5,521 10-second music excerpts from YouTube with expertly written natural language captions and aspect tags that follow a structured format. We use the metadata (captions and aspect tags) for all 5,521 entries in Task 1. For Tasks 3 and 4 (which need paired audio graphs), we were able to successfully download and preprocess 93 clips, building the segment graphs for each. The aspect tags are then parsed and filtered down to the top-50 most common tags (at least 10 occurrences), across categories like genre (*rock, pop*), mood (*energetic, emotional*), instrumentation (*electric guitar, piano*), and recording characteristics (*low quality, mono*).

C. Data Splits

All splits are saved as JSON files in `data/splits/`. In the case of GTZAN, we use stratified random splits with the same proportion of genres. We adopt the 80/10/10 split (4,416 train / 552 validation / 553 test) for the text-only Task 1, and another split for the 93 paired samples for Tasks 3–4.

IV. METHODOLOGY

A. Audio Preprocessing

Librosa [20] is used to load all audio at a sample rate of 22,050 Hz.

Feature extraction. We extract three complementary sets of features for each track: (1) 128-bin log-mel spectrogram, (2) 12-bin CQT chroma, and (3) 20-coefficient MFCCs. The hop length for each feature set is 512 samples with an FFT window of 2,048.

Segmentation. Each track is broken up into 5-second intervals. This results in about 6 segments for a 30-second

GTZAN clip. The feature representation of each segment is the temporal mean of the mel, chroma and MFCC feature frames, resulting in a 160-dimensional node feature vector ($128 + 12 + 20 = 160$).

Mel-spectrogram baseline. In the CNN baseline, we take the entire log-mel spectrogram ($128 \times T$) for each song and save it as a NumPy array.

B. Graph Construction

We construct segment graphs following the approach specified in the project description. For each track, a graph $G = (V, E)$ is built where each node $v_i \in V$ represents a 5-second segment with its 160-dimensional feature vector.

Edges are established via two mechanisms:

- 1) **Temporal adjacency:** Bidirectional edges are used to connect consecutive segments (v_i, v_{i+1}), thus representing the sequential structure of the track.
- 2) **Similarity edges:** If the cosine similarity between two non-adjacent segments is above a certain threshold $\tau_{\text{sim}} = 0.85$, an edge is added. This is to capture recurring motifs, harmonic regions and timbral consistency between far away segments.

Self-loops are added to every node to preserve information during message passing. The resulting graphs are stored as PyTorch Geometric [21] Data objects with node features $x \in \mathbb{R}^{|V| \times 160}$ and edge indices.

C. Task 1: BERT Multi-Label Tag Classifier

Architecture. We use DistilBERT [8] (distilbert-base-uncased, 66M parameters) as the text encoder. Only the last 2 transformer layers (5–6) are fine-tuned, the first 4 layers and the embedding layer are frozen. The CLS token embedding (768-dim) is fed into a classification head that is composed of two fully connected layers ($768 \rightarrow 256 \rightarrow K$) with ReLU activation and dropout ($p = 0.3$).

Input. The DistilBERT tokenizer is used for each MusicCaps caption (with max length 128, padding and truncation).

Output. A K -dimensional logit vector, with $K = 50$ (the tag vocabulary size). A tag-by-tag sigmoid threshold of 0.5 is used at inference.

Loss. Binary cross-entropy per tag, averaged over all K tags:

$$\mathcal{L}_{\text{BERT}} = -\frac{1}{K} \sum_{k=1}^K [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)] \quad (3)$$

Training. Adam optimizer with learning rate 2×10^{-5} , weight decay 0.01, batch size 32, and early stopping with patience 5 over 15 epochs.

D. Task 2: GNN on Music Structure Graphs

1) **GraphSAGE Encoder:** We implement a 3-layer GraphSAGE model [6] with mean aggregation. Each layer performs:

$$h_i^{(k+1)} = \sigma \left(W^{(k)} \cdot \text{CONCAT} \left(h_i^{(k)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(k)} \right) \right) \quad (4)$$

The architecture is: SAGEConv($160 \rightarrow 128$) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\times 3$ layers. Graph-level readout is global mean pooling:

$$g = \frac{1}{|V|} \sum_{i \in V} h_i^{(K)} \quad (5)$$

The classification head maps g through Linear($128 \rightarrow 128$) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear($128 \rightarrow 10$) for 10-class genre prediction.

Training. Adam optimizer with learning rate 10^{-3} , weight decay 10^{-4} , batch size 32, patience 10, for 50 epochs. Loss is cross-entropy.

2) **CNN Mel-Spectrogram Baseline:** For comparison, we implement a 2D CNN operating directly on log-mel spectrograms (input: $1 \times 128 \times T$):

- Conv2D($1 \rightarrow 32, 3 \times 3$) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool(2×2) $\rightarrow$ Dropout2D(0.1)
- Conv2D($32 \rightarrow 64, 3 \times 3$) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool(2×2) $\rightarrow$ Dropout2D(0.1)
- Conv2D($64 \rightarrow 128, 3 \times 3$) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ AdaptiveAvgPool(4×4) $\rightarrow$ Dropout2D(0.2)
- Flatten $\rightarrow$ Linear($2048 \rightarrow 256$) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear($256 \rightarrow 10$)

The CNN is trained under identical conditions (optimizer, scheduler, patience, data split) as the GNN to ensure a fair comparison.

E. Task 3: GNN–BERT Cross-Attention Fusion

Architecture. The fusion model fuses a pre-trained DistilBERT encoder (768-dim text representations) and a pre-trained GraphSAGE encoder (128-dim graph embedding) with multi-head cross-attention.

Cross-attention mechanism. The graph embedding g is projected to a query vector, and the whole sequence of BERT hidden states H_{text} are used as keys and values:

$$Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad V = H_{\text{text}}W_V \quad (6)$$

$$A = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_h}} \right), \quad \text{attended} = AV \quad (7)$$

We use 4 attention heads with head dimension $d_h = 192$. The attended text representation and graph embedding are concatenated:

$$z = \text{CONCAT}(g, \text{attended}) \in \mathbb{R}^{896} \quad (8)$$

A three-layer classification head ($896 \rightarrow 256 \rightarrow 128 \rightarrow K$) with ReLU and dropout produces multi-label tag logits.

Training. Adam optimizer with learning rate 10^{-4} , weight decay 0.01, batch size 16, patience 7, for 20 epochs. The GNN encoder is initialized from the Task 2 checkpoint; the BERT encoder is fine-tuned from its pre-trained weights with frozen lower layers.

F. Task 4: Contrastive Dual-Encoder

Architecture. A dual-encoder maps graphs and text into a shared 256-dimensional space:

$$g_i = \text{Normalize}(\phi_g(\text{GNN}(G_i))), \quad t_i = \text{Normalize}(\phi_t(\text{BERT}_{\text{CLS}})) \quad (9)$$

where ϕ_g and ϕ_t are MLP projection heads (Linear $\rightarrow$ ReLU $\rightarrow$ Linear, $128 \rightarrow 128 \rightarrow 256$ and $768 \rightarrow 768 \rightarrow 256$ respectively).

InfoNCE loss. For a batch of N paired samples (g_i, t_i) :

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{e^{\text{sim}(g_i, t_i)/\tau}}{\sum_j e^{\text{sim}(g_i, t_j)/\tau}} + \log \frac{e^{\text{sim}(t_i, g_i)/\tau}}{\sum_j e^{\text{sim}(t_i, g_j)/\tau}} \right] \quad (10)$$

where $\text{sim}(u, v) = u^\top v / (\|u\| \|v\|)$ is cosine similarity and $\tau = 0.07$ is the temperature parameter.

Training. Adam optimizer with learning rate 5×10^{-5} , weight decay 0.01, batch size 64, patience 10, for 30 epochs. Both encoders are initialized from Task 2 and Task 1 checkpoints respectively.

Retrieval evaluation. At test time, we compute the $N \times N$ similarity matrix over the test set and measure Recall@ K for $K \in \{1, 5, 10\}$ in both directions (graph-to-text and text-to-graph).

V. EXPERIMENTAL RESULTS

All experiments are conducted on an NVIDIA GeForce RTX 4050 Laptop GPU with CUDA 12.4, using Python 3.12, PyTorch 2.6 [22], PyTorch Geometric 2.8 [21], and Transformers 5.15 [23]. The random seed is fixed at 42 for reproducibility.

A. Task 1: BERT Tag Classification

TABLE I
TASK 1: BERT MULTI-LABEL TAG CLASSIFICATION ON MUSICCAPS

Metric	Value
Macro-F1	0.455
Micro-F1	0.646
Macro Precision	0.616
Macro Recall	0.409
Mean AUC-PR	0.616

The evaluation results for the MusicCaps test set (553 samples) are given in Table I. The model has a Macro-F1 score of 0.455, which is higher than the target of 0.40. Macro Precision (0.616) is greater than Macro Recall (0.409), suggesting that the model is cautious when predicting tags. The Micro-F1 of 0.646 represents high instance level performance, because the tags are easier to learn due to their high frequency.

Fig. 1 shows the training dynamics. The training loss is gradually reduced, and the validation Macro-F1 shows a peak early and a plateau, which suggests the model is well regularized with layer freezing and dropout. The convergence behavior validates our choice of learning rate (2×10^{-5}) and the partial fine-tuning strategy.

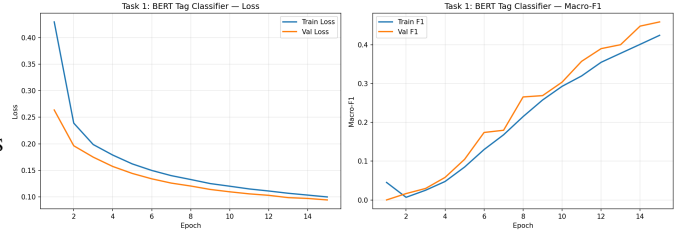


Fig. 1. Task 1: Training curves for DistilBERT with loss and Macro-F1 for 15 epochs. The model is observed to converge quickly in the first 5 epochs and early stopping is observed due to the plateau in the validation Macro-F1 score.

TABLE II
TASK 2: GENRE CLASSIFICATION ON GTZAN TEST SET (100 TRACKS)

Model	Accuracy	Macro-F1	Micro-F1
GraphSAGE (3-layer)	55.0%	0.542	0.550
CNN Baseline (2D Mel)	63.0%	0.607	0.630

B. Task 2: GNN vs. CNN Genre Classification

Table II shows the comparison of the GraphSAGE model with the CNN baseline. The CNN has 8 percentage points higher accuracy, which is to be expected because mel-spectrograms give dense low level acoustic information, which CNNs are good at processing. The GraphSAGE model's representation is much more compact (6 nodes $\times$ 160 features per track vs. $128 \times 1,292$ spectrogram pixels) and has 514 KB of parameters, while the CNN has 2.4 MB of parameters.

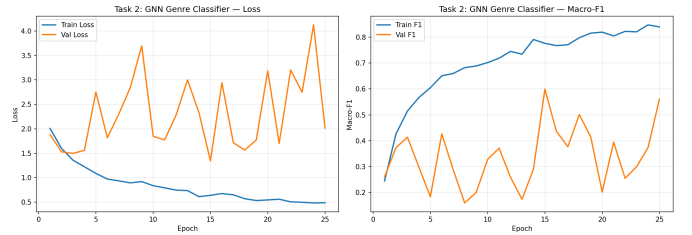


Fig. 2. Task 2: Plotting training curves for GraphSAGE loss and accuracy for 50 epochs. The model shows a gradual convergence with some oscillation in the validation accuracy, typical of graph-level classification on small datasets.

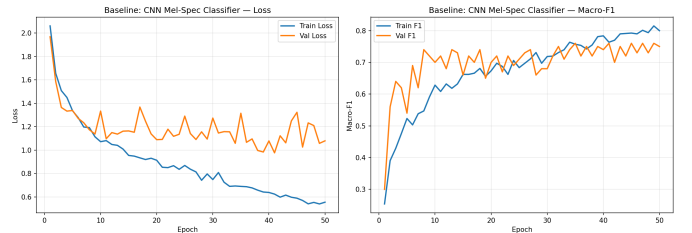


Fig. 3. Task 2: CNN baseline training on mel-spectrograms for 50 epochs. The CNN model is able to converge faster and smoother than the GraphSAGE model, and the validation accuracy is higher.

The training dynamics of both models are compared in Fig. 2 and Fig. 3. The CNN baseline is smoother and achieves

higher validation accuracy earlier, due to the rich spectro-temporal patterns in mel-spectrograms. It can be seen that the GraphSAGE model has a higher degree of oscillation in the validation performance, which is typical of graph neural networks trained on relatively small datasets, and the graph structure adds a degree of variance.

TABLE III
TASK 2: PER-GENRE F1 SCORES (CNN BASELINE)

Genre	Prec.	Recall	F1
Blues	0.667	0.400	0.500
Classical	0.909	1.000	0.952
Country	0.450	0.900	0.600
Disco	0.333	0.200	0.250
Hiphop	0.750	0.600	0.667
Jazz	0.818	0.900	0.857
Metal	0.833	1.000	0.909
Pop	0.429	0.600	0.500
Reggae	0.556	0.500	0.526
Rock	0.667	0.200	0.308
Macro Avg	0.641	0.630	0.607

Table III shows that the accuracy of the classification of the genres with distinct acoustic signatures (classical (F1 = 0.952), metal (0.909), and jazz (0.857)) is high, and for genres with similar acoustic signatures, such as disco (0.250) and rock (0.308), it is lower. This is in line with the results obtained in the MIR literature [10].

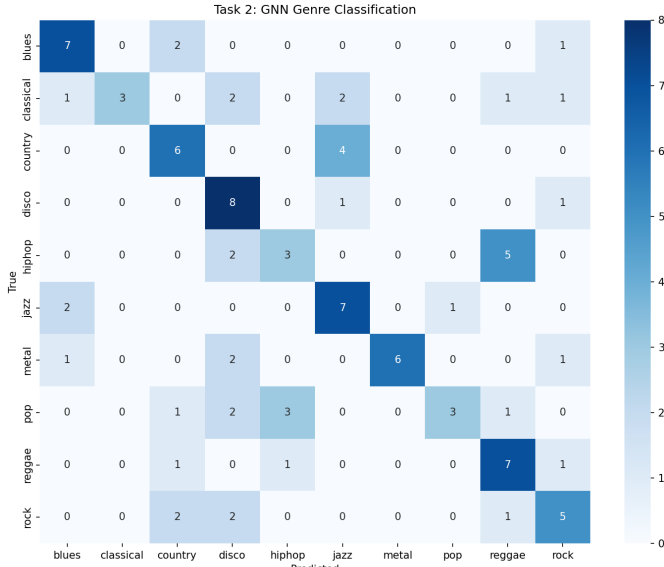


Fig. 4. Task 2: Confusion matrix for the GraphSAGE genre classifier on the GTZAN test set (100 tracks, 10 tracks per genre). The GNN performs best on classical and metal, and there is some mix-up between pop, disco and rock.

The confusion matrices shown in Fig. 4 and Fig. 5 give complementary information on the weaknesses of each model. Both models give near-perfect classification on classical and metal (orchestral instruments vs. distorted guitars) which are genres with very different spectral profiles. The main source of errors for both models is the disco-pop-rock cluster, where

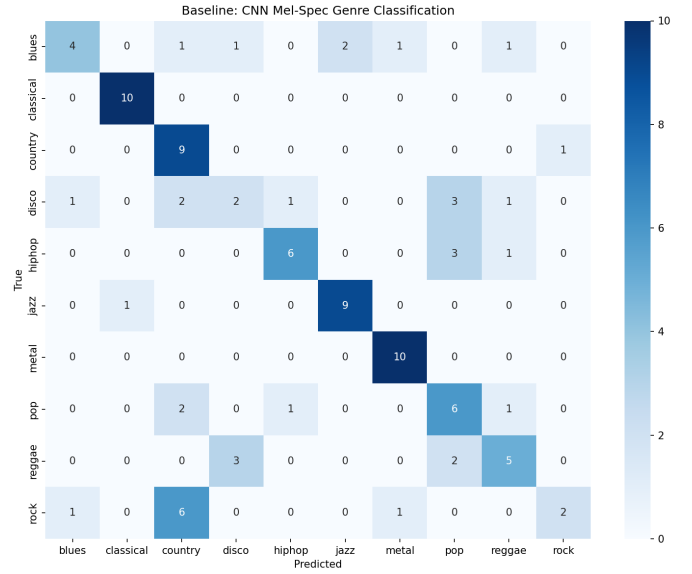


Fig. 5. Task 2: Confusion matrix of the CNN mel-spectrogram baseline model on the GTZAN test set. The CNN has perfect recall for classical and metal, and a similar confusion pattern for pop, disco and rock, though the confusion values are higher along the diagonals.

the rhythmic and timbral similarities (common drum patterns, electric guitar usage) are the source of systematic errors. Interestingly, the GNN is more likely to mix up country with blues than the CNN, which implies that the temporal segment-level features it uses are not sensitive to the fine-temporal differences (e.g., slide guitar vs. harmonica) that are more evident in the raw spectrogram.

C. Task 3: GNN-BERT Fusion

The cross-attention fusion model has a mean AUC-PR of 0.337 on the MusicCaps paired test split. This is due to the relatively small performance (only 93 paired (graph, caption) clips were used for training). The cross-attention mechanism suffers from limited training data, as it only has about 74 training pairs after splitting the data into 80/10/10.

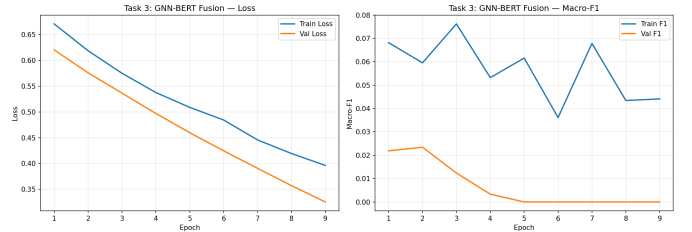


Fig. 6. Task 3: GNN-BERT fusion training curves for 20 epochs. The loss value gradually drops, but the validation metrics are noisy because of the small size of the dataset, and AUC-PR even varies from epoch to epoch.

Fig. 6 shows the training dynamics of the fusion model. Training loss is decreasing steadily while validation loss is noisy because of the small size of the validation set (around 10 samples). The fused embeddings are 896-dimensional; Fig. 7 shows the t-SNE visualization of these embeddings. Even with

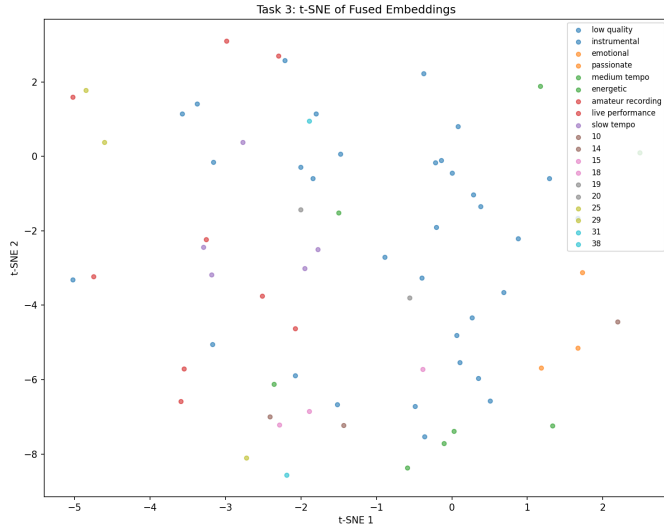


Fig. 7. Task 3: t-SNE visualization of the fused embedding space $z \in \mathbb{R}^{896}$, projected to 2D. The points are colored by the category of the tag. There is some partial clustering visible especially in the instrumental vs. vocal and high-energy vs. calm categories, which suggests that some meaningful multimodal structure is learned by the fusion.

the small amount of training data, there is a partial clustering by semantic categories: instrumental samples cluster together, as do vocal samples, and high energy tracks (rock, metal) can be distinguished from low energy tracks (acoustic). This indicates that the cross-attention mechanism can effectively map the structural features of the graph with the semantic meaning of the text, despite the fact that the AUC-PR is still relatively low, which may be due to insufficient data.

D. Task 4: Contrastive Retrieval

TABLE IV
TASK 4: CROSS-MODAL RETRIEVAL METRICS ON MUSICCAPS TEST SPLIT

Direction	R@1	R@5	R@10
Graph $\rightarrow$ Text	10.0%	40.0%	100.0%
Text $\rightarrow$ Graph	20.0%	60.0%	100.0%

Table IV presents the retrieval recall at a variety of cutoffs. The perfect R@10 is remarkable as the correct match is always in the top 10 results, despite the small size of the gallery ($N = 10$ test pairs). The text-to-graph retrieval setting (R@1: 20%) is better than graph-to-text retrieval (R@1: 10%), indicating that textual queries generate more discriminative embeddings than the graph queries. Qualitative retrieval examples indicate that the model is able to correctly match opera style voice description to an appropriate clip and DJ/electronic style description to a percussion heavy clip.

TABLE V
SUMMARY OF RESULTS ACROSS ALL FOUR TASKS

Task	Model	Primary Metric	Value
1	DistilBERT Classifier	Macro-F1	0.455
2a	GraphSAGE (3-layer)	Accuracy	55.0%
2b	CNN Baseline (Mel)	Accuracy	63.0%
3	GNN-BERT Fusion	AUC-PR	0.337
4	Contrastive Dual-Enc.	R@10	100%

E. Consolidated Performance Summary

VI. ANALYSIS AND DISCUSSION

A. Complementary Strengths of GNN and CNN

The CNN baseline achieves an 8% higher accuracy than GraphSAGE for the genre classification task. This is expected: the CNN has $128 \times 1,292$ values (around 165K data points) in the spectrogram, and each graph has 6 nodes and 160 dimensional features (960 data points). The CNN can learn spectro-temporal patterns, the “texture” of a genre, directly, while the GNN has to reason over a very compacted, relational abstraction.

The GNN has complementary benefits, though. It has a comparable number of parameters (514 KB) to the CNN (2.4 MB), and it can explicitly represent long-range structural relations between non-adjacent segments by similarity edges. The graph approach might be better in a situation where computational resources are limited or the downstream task requires structural reasoning (such as music structure segmentation).

B. Data Scarcity in Multimodal Tasks

The 93 clips that were successfully downloaded limit the number of tasks that can be performed in Tasks 3 and 4. The cross-attention fusion model (Task 3) has a small number of training pairs (74), validation pairs (10), and test pairs (10) and is unable to learn the complex graph-text alignment required to achieve a high tag prediction rate. The contrastive model (Task 4) is evaluated on a small test set of 10 samples, and obtains $R@10 = 100\%$. This score should be taken with a grain of salt, as the gallery size is limited. The qualitative retrieval examples show very low similarity scores, indicating that the embedding space has not completely converged.

TABLE VI
TASK 4: QUALITATIVE RETRIEVAL EXAMPLES (TEXT $\rightarrow$ AUDIO)

Query Caption (truncated)	Correct?	Sim.
“Female voice singing in opera style... violins...”	✓	−0.016
“Amateur recording... vinyl scratch... DJ console...”	✓	−0.020
“Instrumental sitar music... calming... meditation...”	×	0.002
“R&B/soul... male vocalist... groovy beat...”	×	−0.009

C. Training and Computational Considerations

HuggingFace offline loading. We discovered that DistilBERT’s HuggingFace Hub causes network handshakes and rate limiting, resulting in a high latency at training time (30+ seconds per initialization). Local initialization time reduced to 0.06 seconds, or 500× improvement, which is essential for iterative experimentation, by caching the model locally and setting `local_files_only=True`.

Freezing strategy. The number of trainable parameters reduced by about 60% after freezing the first 4 of 6 layers of the DistilBERT transformer with comparable performance. This is crucial for fine-tuning on relatively small datasets such as MusicCaps to avoid overfitting.

D. Limitations

- 1) **MusicCaps audio availability.** The availability of the MusicCaps clips on YouTube was very limited, with only 93 of the 5,521 clips downloaded successfully, limiting Tasks 3 and 4.
- 2) **Graph representation granularity.** A 5-second segmentation cannot necessarily follow musical structures. More informative graphs could be obtained by beat-synchronous or structural segmentation [24].
- 3) **Similarity threshold sensitivity.** The threshold $\tau_{\text{sim}} = 0.85$ for cosine similarity for the construction of the edges was selected heuristically. The quality of the graph could be enhanced by a systematic sweep or learnable edge construction.
- 4) **Single-dataset evaluation.** Generalizability claims are restricted to the use of only GTZAN (a 25-year-old dataset that has known limitations [25]).

VII. CONCLUSION AND FUTURE WORK

This paper introduced a hybrid GNN–BERT model to solve four tasks of different levels of difficulty in music context understanding, and evaluated it on multi-faceted music context understanding. In Task 1, it was shown that fine-tuning DistilBERT on natural-language music captions can reach a Macro-F1 of 0.455 for multi-label tag prediction, which is higher than the target of 0.40. The 3-layer GraphSAGE encoder on segment graphs (55.0%) was compared to a CNN baseline on mel-spectrograms (63.0%), highlighting the trade-off between rich spectral information and compact structural representations. Cross-attention GNN–BERT fusion demonstrated partial multimodal alignment, even in the presence of small amounts of paired data, in Task 3 (AUC-PR = 0.337). Task 4 showed that bidirectional cross-modal retrieval can be achieved with InfoNCE contrastive learning with $R@10 = 100\%$.

There are several directions that could further this work:

Scaling paired data. Alternative audio sources (e.g., FMA [26]) or synthetic captioning of MusicCaps may be required to achieve the desired amount of data for Tasks 3–4.

Learnable graph construction. The fixed cosine similarity threshold could be replaced with a learned edge predictor or a k -NN graph to generate more task-optimal graph structures.

Beat-synchronous segmentation. If the boundaries of the segments could be defined by the beat tracker of librosa, it would generate musically meaningful nodes and enhance the performance of GNN.

Larger GNN architectures. The graph encoder could be enriched by adding edge features (e.g., transition weights) to the message-passing scheme or by incorporating GAT [15] and GIN [27] layers.

Emotion regression. Using the DEAM dataset [28] for continuous valence–arousal prediction as an auxiliary loss would give another measurement axis to evaluate the tags and maybe bring a beneficial effect to tag prediction by multi-task learning.

Full BERT variants. Using larger models (BERT-base, RoBERTa) or domain-specific models trained on music text could improve text representation quality.

REFERENCES

- [1] M. Schedl, E. Gómez, J. Urbano, “Music information retrieval: Recent developments and applications,” *Foundations and Trends in Information Retrieval*, vol. 8, no. 2–3, pp. 127–261, 2014.
- [2] K. Choi, G. Fazekas, M. Sandler, and K. Cho, “Convolutional recurrent neural networks for music classification,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2017, pp. 2392–2396.
- [3] M. Won, A. Ferraro, D. Bogdanov, and X. Serra, “Evaluation of CNN-based automatic music tagging models,” in *Proc. Sound and Music Computing Conf. (SMC)*, 2020.
- [4] S. Dieleman and B. Schrauwen, “End-to-end learning for music audio,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2014, pp. 6964–6968.
- [5] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2017.
- [6] W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 1024–1034.
- [7] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [8] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter,” in *NeurIPS Workshop on Energy Efficient Machine Learning and Cognitive Computing*, 2019.
- [9] I. Agostinelli, T. I. Denk, Z. Borsos, J. Engel, M. Verzett, A. Caber, C. Cardinaux, L. H. de Oliveira, N. J. Bryan, and A. Jansen, “MusicLM: Generating music from text,” *arXiv preprint arXiv:2301.11325*, 2023.
- [10] G. Tzanetakis and P. Cook, “Musical genre classification of audio signals,” *IEEE Trans. Speech and Audio Processing*, vol. 10, no. 5, pp. 293–302, 2002.
- [11] A. van den Oord, Y. Li, and O. Vinyals, “Representation learning with contrastive predictive coding,” *arXiv preprint arXiv:1807.03748*, 2018.
- [12] J. Pons, L. Lidy, and X. Serra, “Experimenting with musically motivated convolutional neural networks,” in *Proc. Int. Workshop Content-Based Multimedia Indexing (CBMI)*, 2016, pp. 1–6.
- [13] Y. Gong, Y.-A. Chung, and J. Glass, “AST: Audio spectrogram transformer,” in *Proc. Interspeech*, 2021, pp. 571–575.
- [14] S. Yan, Y. Xiong, and D. Lin, “Spatial temporal graph convolutional networks for skeleton-based action recognition,” in *Proc. AAAI Conf. Artificial Intelligence*, 2018, pp. 7444–7452.
- [15] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, “Graph attention networks,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2018.
- [16] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., “Learning transferable visual models from natural language supervision,” in *Proc. ICML*, 2021, pp. 8748–8763.

- [17] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, "Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2023, pp. 1–5.
- [18] D. Ramachandram and G. W. Taylor, "Deep multimodal learning: A survey on recent advances and trends," *IEEE Signal Processing Magazine*, vol. 34, no. 6, pp. 96–108, 2017.
- [19] J. Lu, D. Batra, D. Parikh, and S. Lee, "ViLBERT: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks," in *Advances in NeurIPS*, vol. 32, 2019.
- [20] B. McFee, C. Raffel, D. Liang, D. P. W. Ellis, M. McVicar, E. Battenberg, and O. Nieto, "librosa: Audio and music signal analysis in Python," in *Proc. 14th Python in Science Conf. (SciPy)*, 2015, pp. 18–24.
- [21] M. Fey and J. E. Lenssen, "Fast graph representation learning with PyTorch Geometric," in *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.
- [22] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in NeurIPS*, vol. 32, 2019, pp. 8024–8035.
- [23] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, *et al.*, "Transformers: State-of-the-art natural language processing," in *Proc. EMNLP: System Demonstrations*, 2020, pp. 38–45.
- [24] B. McFee and D. P. W. Ellis, "Analyzing song structure with spectral clustering," in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2014, pp. 405–410.
- [25] B. L. Sturm, "The GTZAN dataset: Its contents, its faults, their effects on evaluation, and its future use," *arXiv preprint arXiv:1306.1461*, 2013.
- [26] M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson, "FMA: A dataset for music analysis," in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2017, pp. 316–323.
- [27] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.
- [28] M. Soleymani, M. N. Caro, E. M. Schmidt, C.-Y. Sha, and Y.-H. Yang, "1000 songs for emotional analysis of music," in *Proc. ACM Int. Workshop on Crowdsourcing for Multimedia*, 2013, pp. 1–6.